

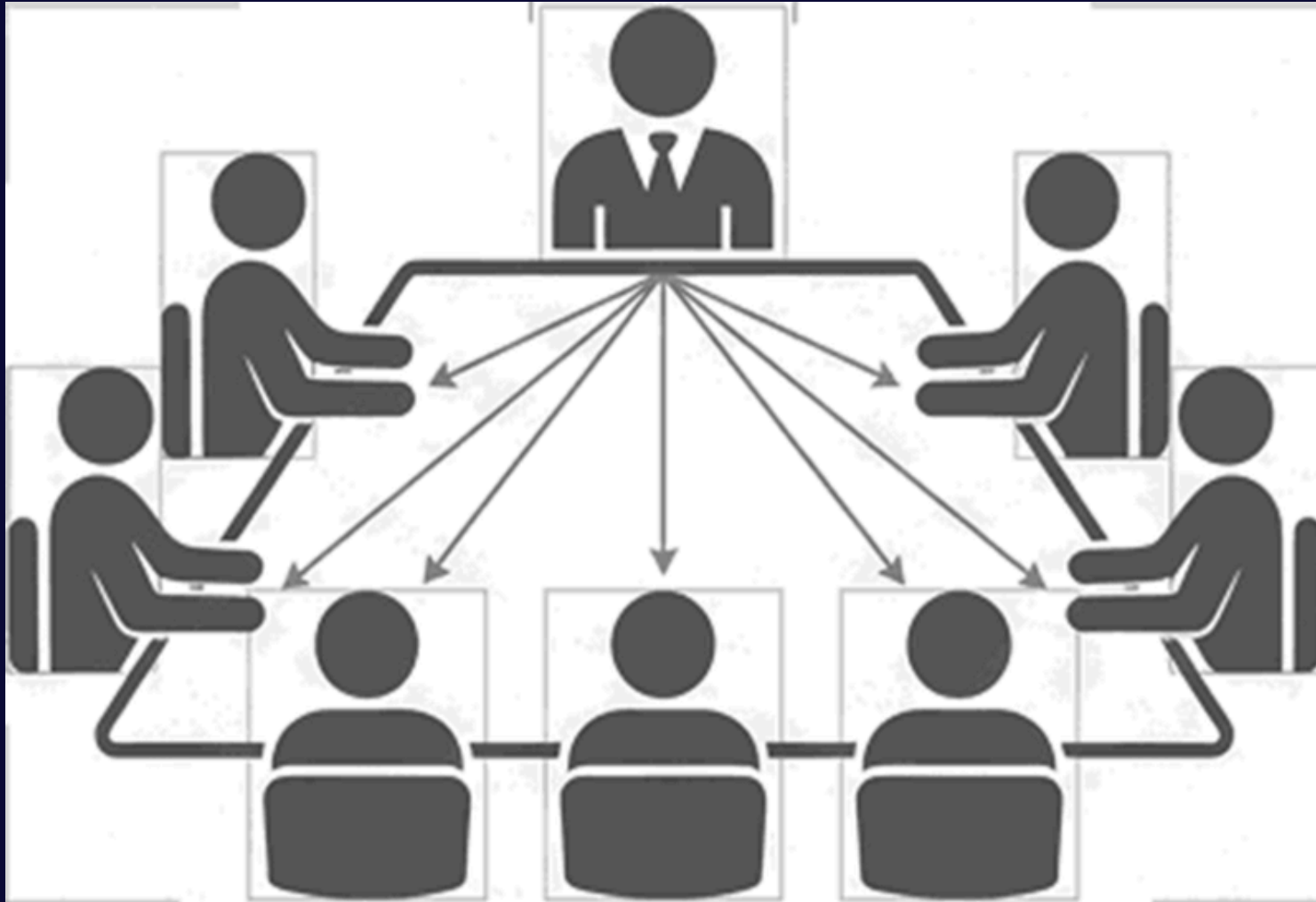
Student Monitoring System

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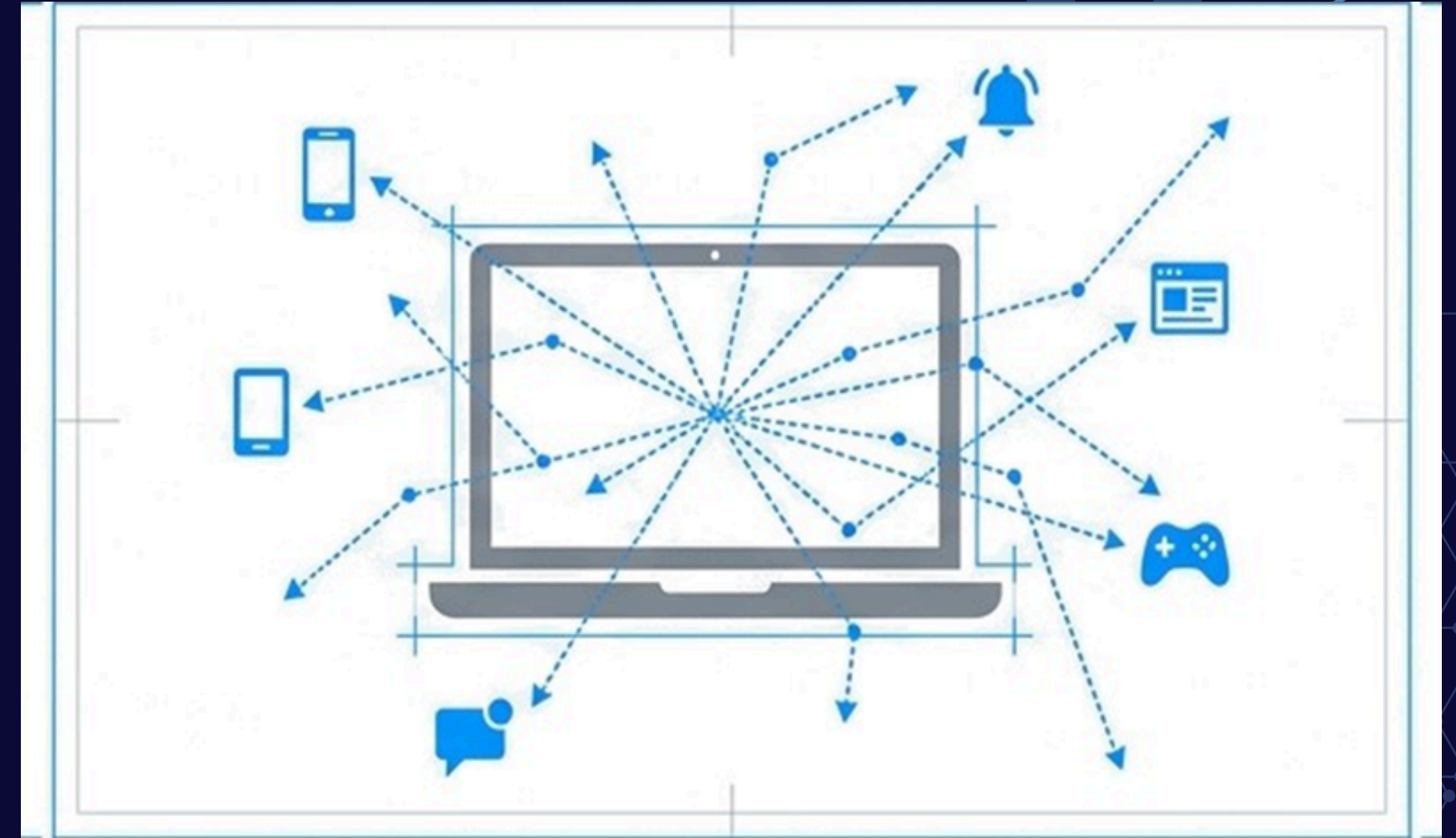
DEPI Graduation Project

2025/2026

Problem : Traditional and Digital Environments



Traditional Environments: Sustained Focus

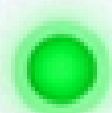


Digital Environments: Fragmented Engagement

The Effect on Engagement



Engagement:
98%



Status:
Active

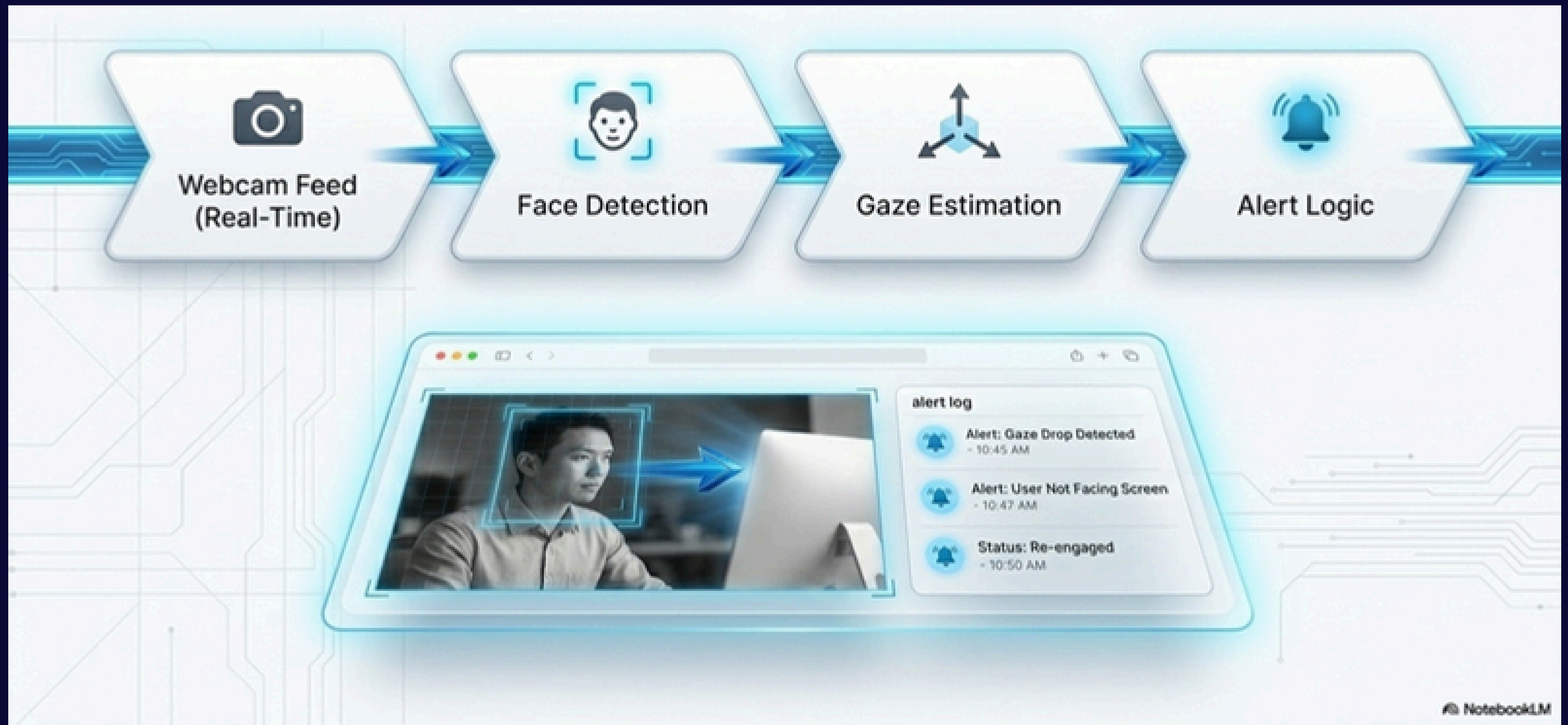


Engagement:
Drop Detected



Alert
Triggered

Solution



DATASET— ETH-XGaze

**a large scale (over
1 million samples)
gaze estimation
dataset**

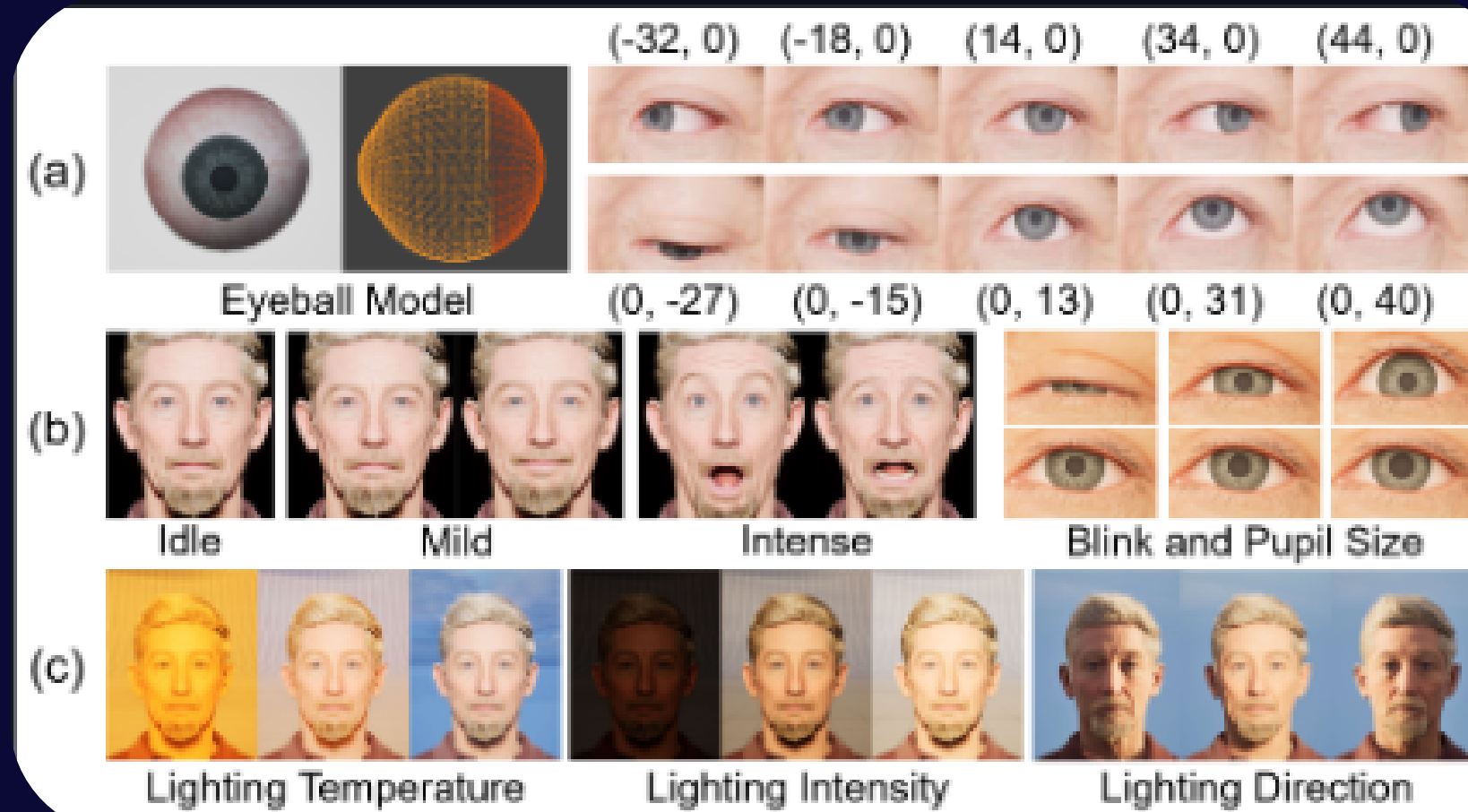


**high-resolution
images under
extreme head poses
and gaze directions**

**110 participants
with a custom
hardware setup**

DATASET— GazeGene

over 1M images with
3D eyeball annotations
and accurate gaze
annotations.



extensive expression, blink
and pupil size, lighting
temperature, lighting
intensity and lighting
direction changes

GazeGene also provides
smooth and large head
pose and gaze distribution
for generalization ability

DATASET— MPIIFaceGaze

**213,659 Total
annotated
images**

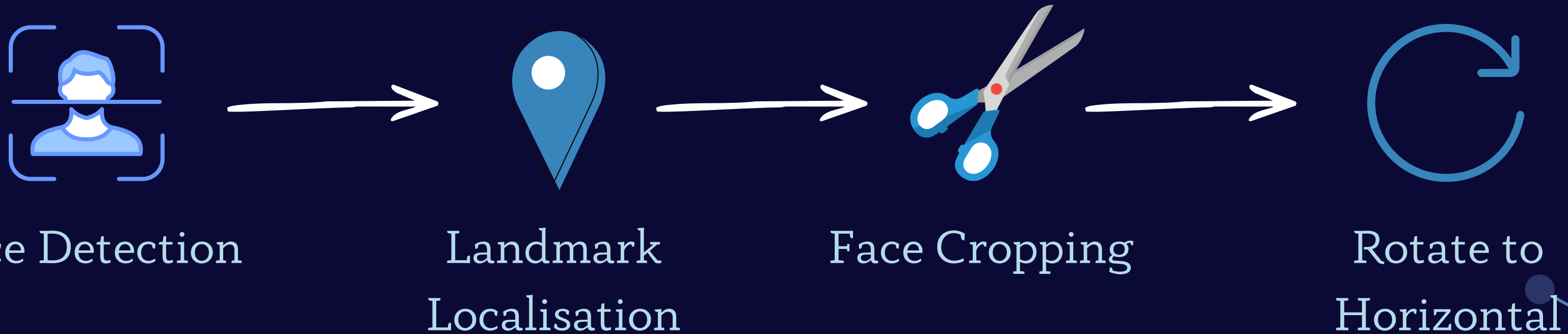
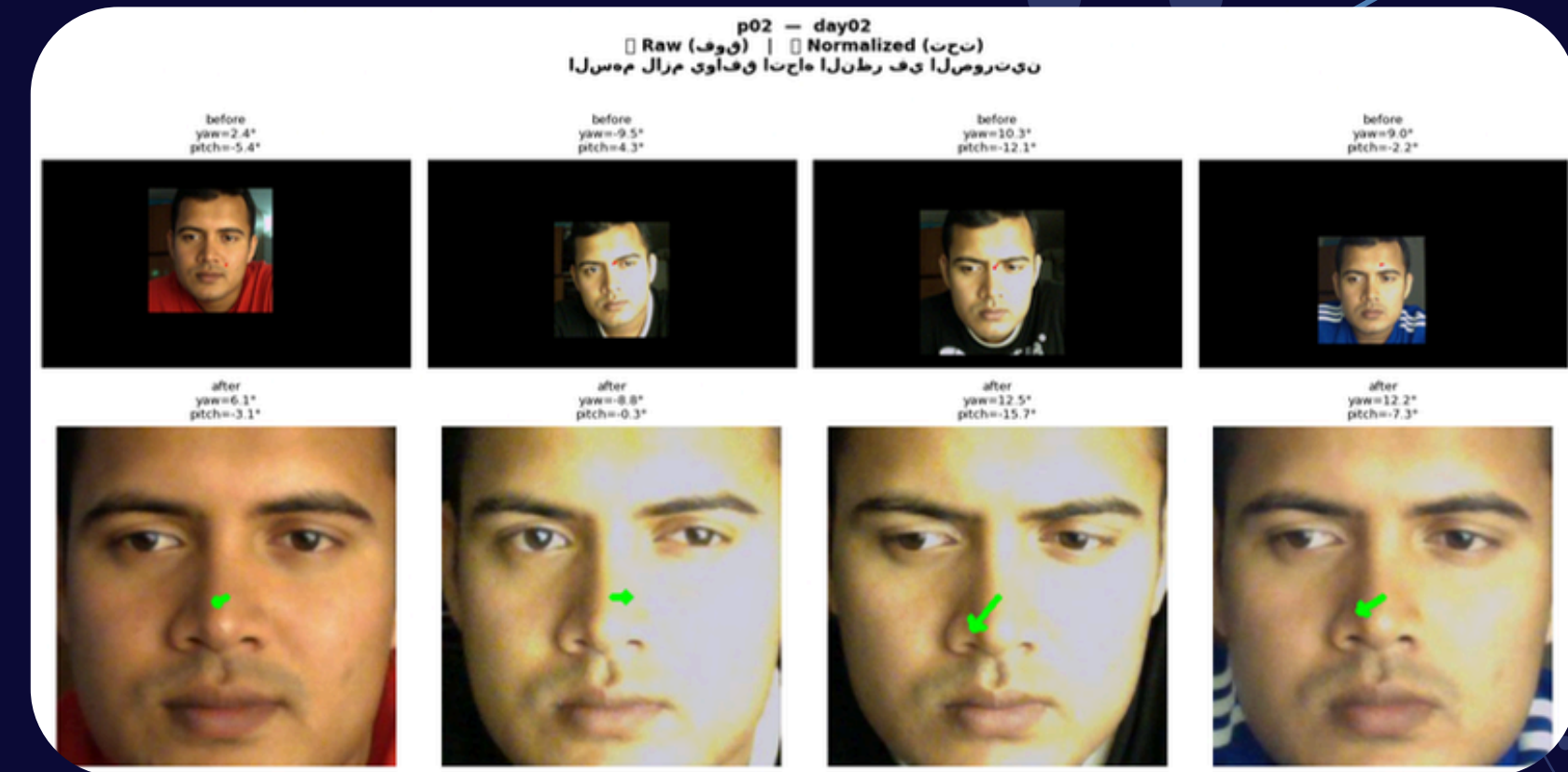
**28 Columns Per
annotation**

**15 Subjects p00 –
p14, collected over
multiple weeks of
natural laptop use**



Top row: raw face images with varied head pose and distance.

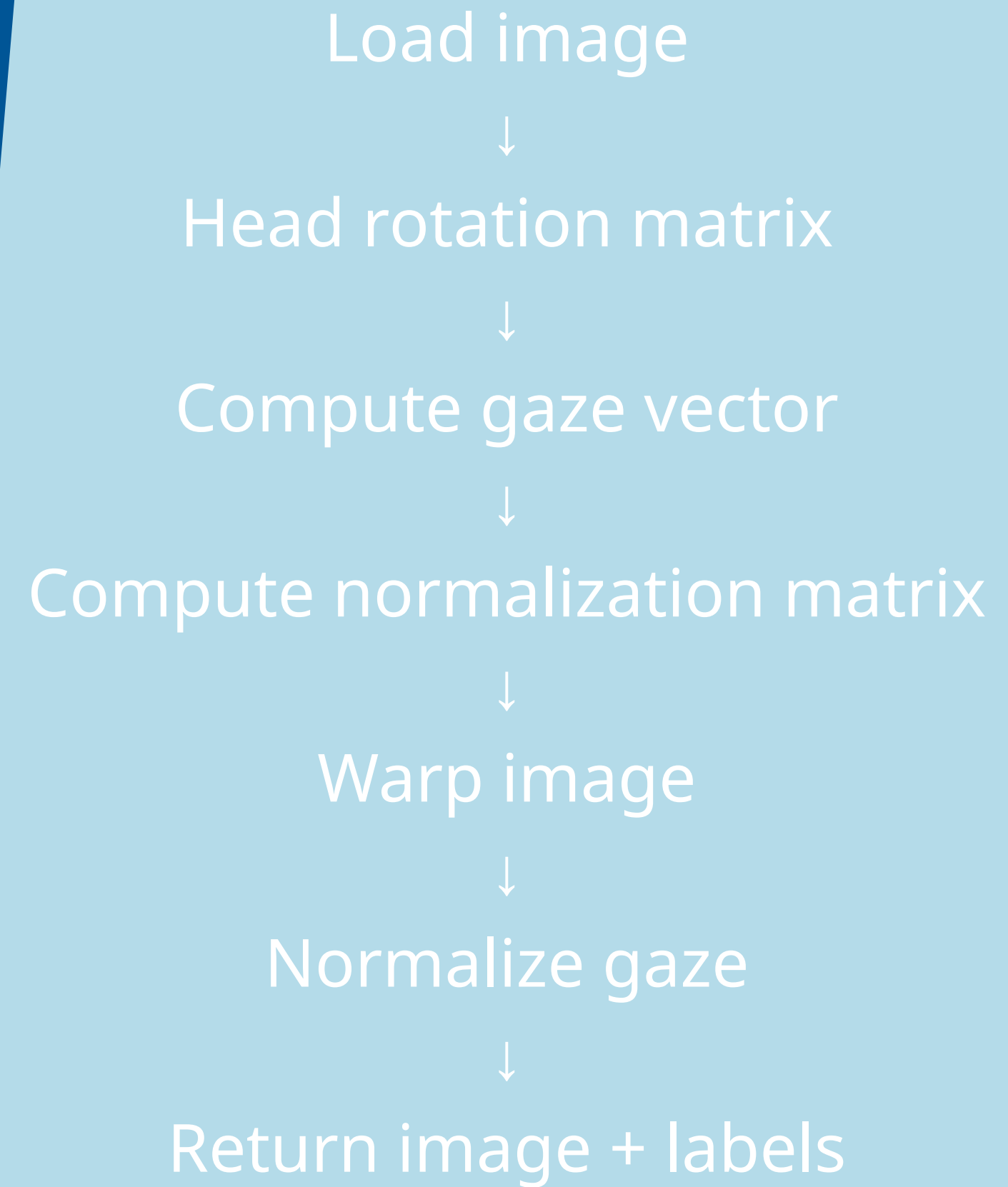
Bottom row: after landmark-guided crop and rotation — the eye patch is aligned horizontally in all cases.



pipeline



09 Presentation

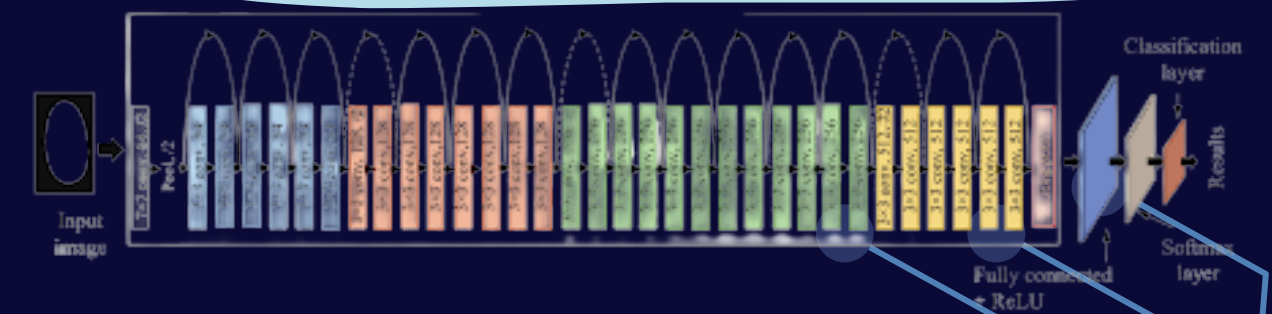
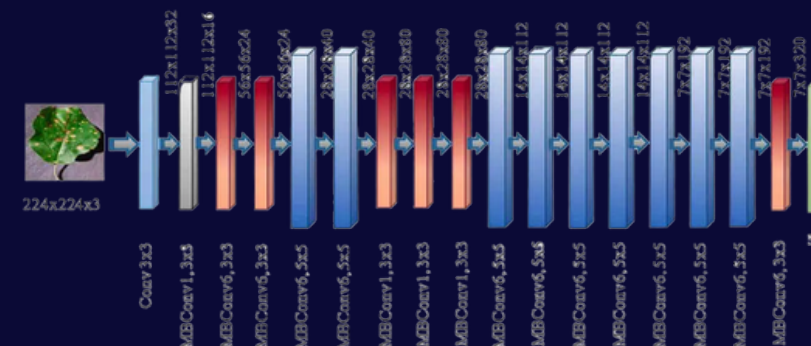


loser

EfficientNB



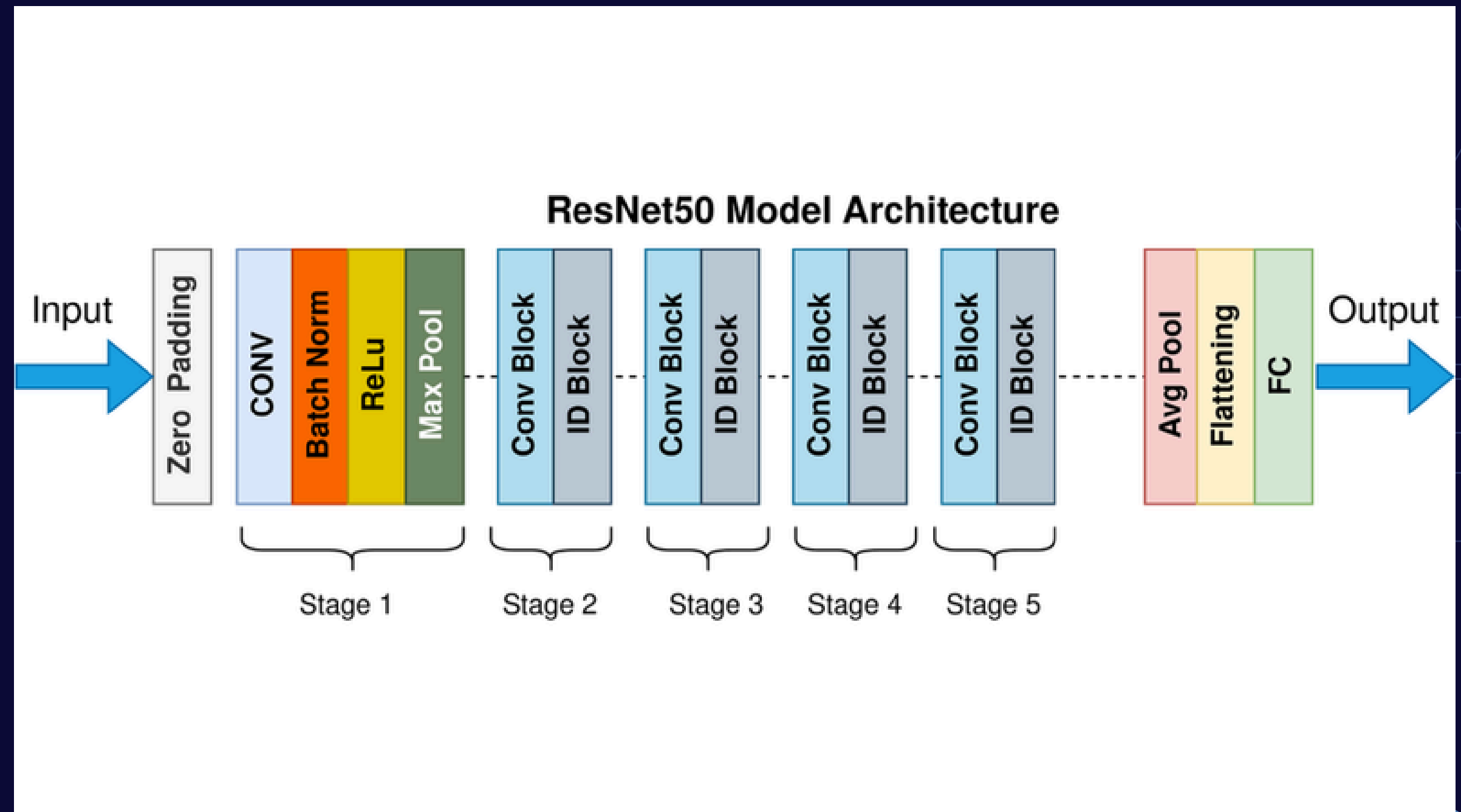
Resnet 50



10 Presentation

Model Architecture

Why ResNet50?
Residual connections
allow training of
deeper networks
**without vanishing
gradients**. Better
accuracy-to-parameter
ratio than VGG16 at
25M parameters.



Traning Techniques

loser



each eye
in parallel
branches

loser



full face + eyes +
head pos

Winner



full face +
head pos

Traning Techniques

Input

processing

Output

full face + pos

ResNet50 (ImageNet
pretrained) ———
ResNet50 (same
weights)

[Yaw (rad), Pitch (rad)]
→ converted to
degrees for
evaluation

Traning Techniques



Dataset
↓
Face Normalization
↓
ResNet50
↓
Predict yaw & pitch
↓
Angular Error Evaluation
↓
Bias Calibration

Training Setup & LOPO

2

TWO-PHASE TRAINING PIPELINE

From Basic Angle Learning to True 3D Gaze Optimization

1 PHASE 1: MAE (SmoothL1) Training

Goal: Learn basic yaw & pitch angles

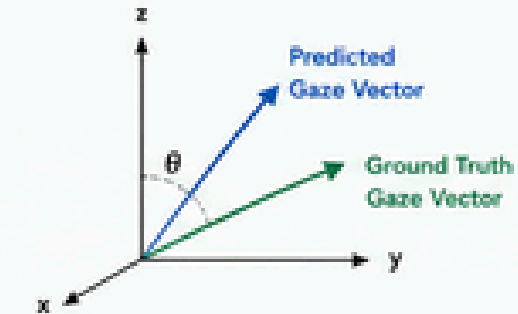
- Loss: SmoothL1 (on yaw and pitch separately)
- Backbone: Frozen for first P1_UNFREEZE epochs (e.g., 5)
- Then: Unfreeze and fine-tune with lower LR



2 PHASE 2: Angular Error Fine-Tuning

Goal: Optimize the model using the true 3D gaze direction

- Loss: Angular Error (3D direction)
- Backbone: Fully trainable
- Fine-tune the whole model with smaller LR



LOSS FUNCTIONS

1. MAEGazeLoss (Phase 1)

SmoothL1 loss on yaw and pitch separately.

$$L = \text{SmoothL1}(\hat{y}_{\text{yaw}} - y_{\text{yaw}}) + \text{SmoothL1}(\hat{y}_{\text{pitch}} - y_{\text{pitch}})$$

Also compute MAE (in degrees) for monitoring:

$$\text{MAE}_{\text{yaw}} = \text{mean}(|\hat{y}_{\text{yaw}} - y_{\text{yaw}}|) \times \frac{180}{\pi}$$

$$\text{MAE}_{\text{pitch}} = \text{mean}(|\hat{y}_{\text{pitch}} - y_{\text{pitch}}|) \times \frac{180}{\pi}$$

2. AngularErrorLoss (Phase 2)

Convert yaw & pitch to 3D gaze vectors and compute the angle between predicted and ground truth vectors.

$$L = \arccos(\hat{g} \cdot g)$$

Where $\hat{g}$ and g are unit 3D gaze vectors.

Yaw & Pitch → 3D Gaze Vector (Used in Phase 2)

$$\begin{aligned}x &= -\cos(\text{pitch}) \cdot \sin(\text{yaw}) \\y &= -\sin(\text{pitch}) \\z &= -\cos(\text{pitch}) \cdot \cos(\text{yaw})\end{aligned}$$

$$g = [x, y, z]^T$$

Angle Between Vectors

$$\cos(\theta) = \hat{g} \cdot g$$

$$\theta = \arccos(\cos(\theta))$$

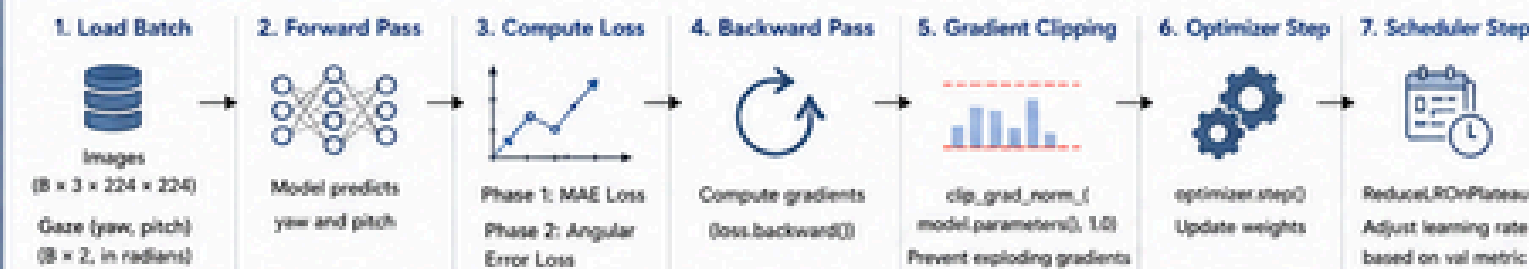
$$-1 + \epsilon \leq \cos(\theta) \leq 1 - \epsilon$$

(for numerical stability)

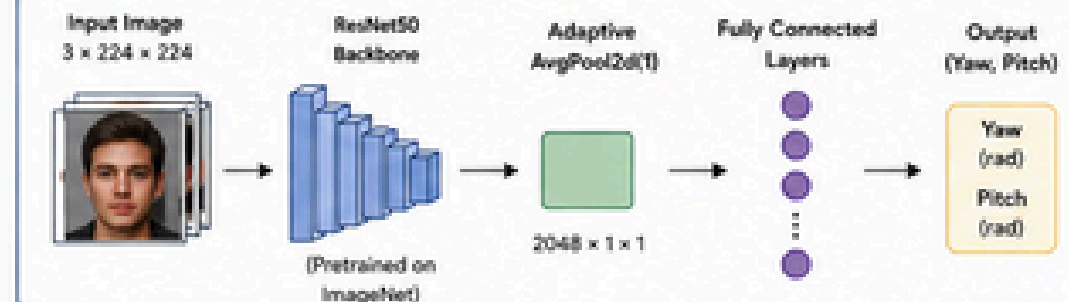
Why Two Phases?

- ✓ Phase 1 is more stable and helps the model learn a good initialization.
- ✓ Phase 2 directly optimizes the true metric (angular error) for best performance.
- ✓ Improves convergence and final accuracy.

TRAINING LOOP (Per Epoch)



MODEL ARCHITECTURE (GazeResNet)



DATA LOADING & AUGMENTATION



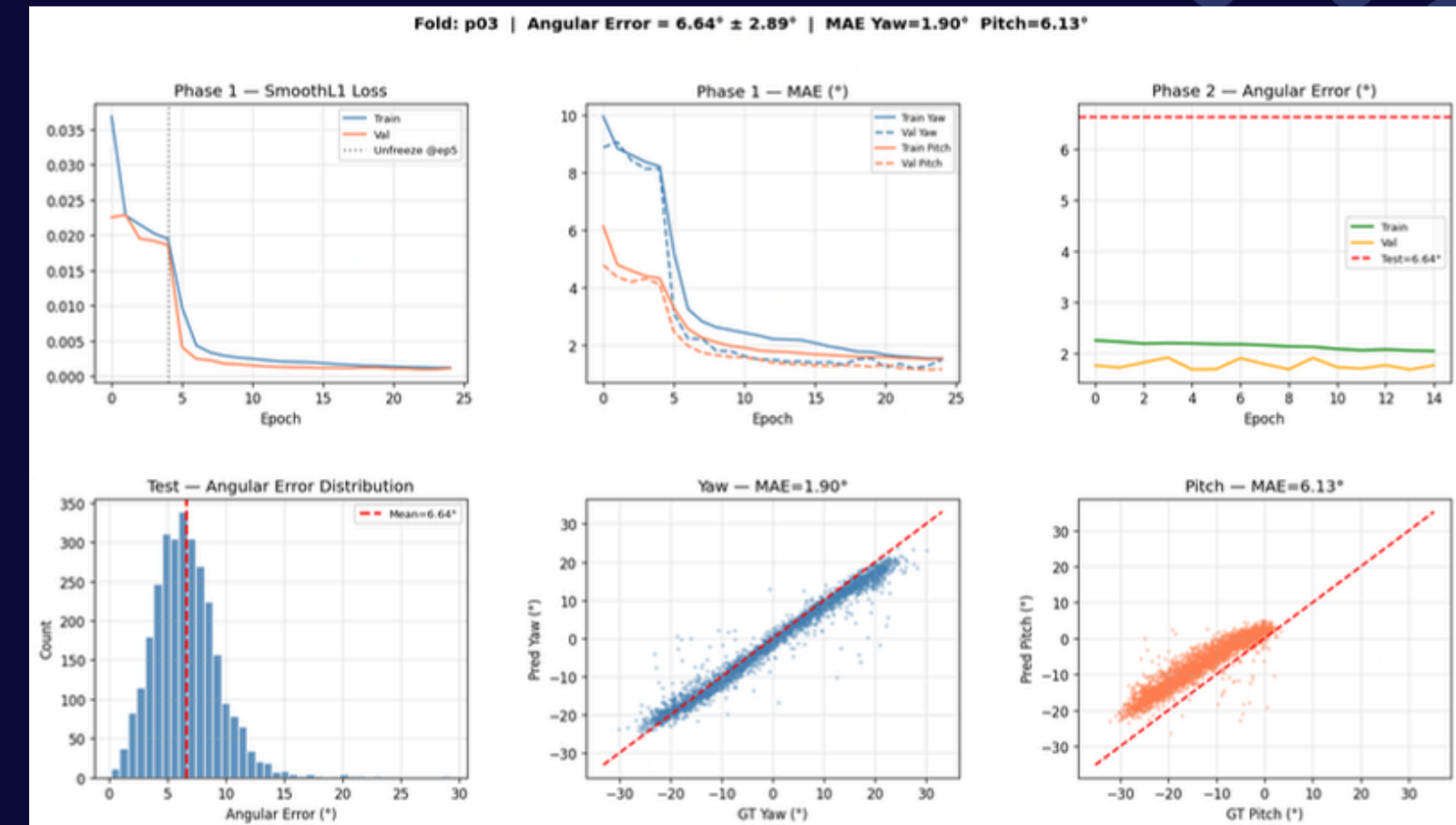
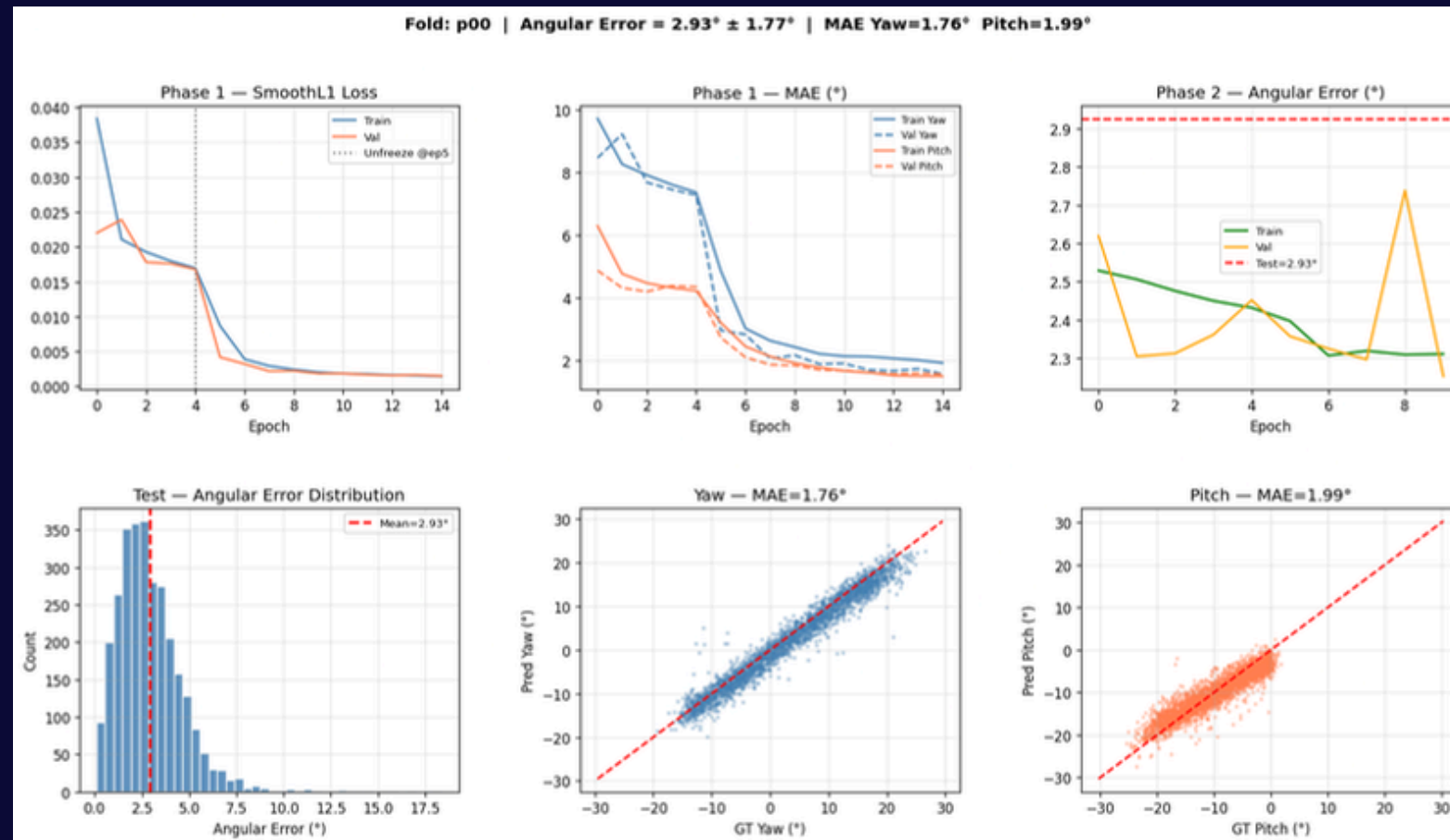
TRAIN / VAL / TEST SPLIT (Per Fold)



EVALUATION METRICS

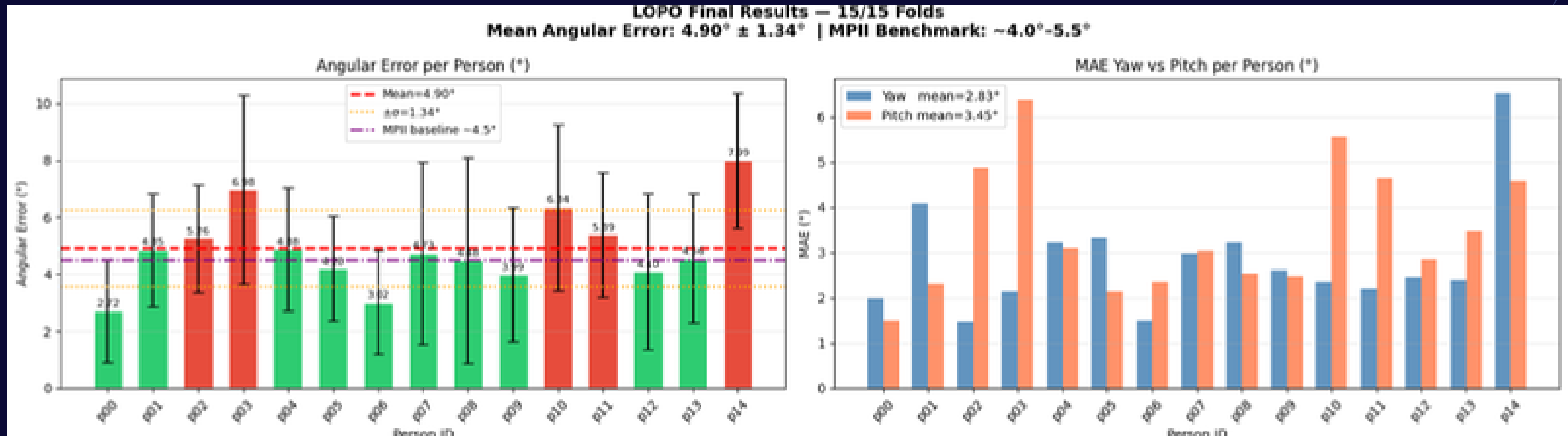
- Angular Error (°)**
Main evaluation metric (3D direction)
- MAE Yaw (°)**
Mean absolute error of yaw
- MAE Pitch (°)**
Mean absolute error of pitch

Results



Per-fold training curves (loss and angular error) and error distributions. Each panel corresponds to one held-out subject.

Results



Challenges



3%



Solution



Per-Session Calibration At the start of each session, the student looks at 4 points displayed on screen — one in each corner.

RESULTS



Yellow circle = predicted gaze location · Red cross = ground truth · Four panels show different gaze directions. The model correctly tracks the general direction in all four cases.

Deployment

What Does This System Do?

Monitors Every Student

Tracks each student's focus level in real time throughout the lecture

Calculates Attention Score

Identifies who is focused and who is distracted at any moment

Instant Alerts

If a student is distracted more than 3 times, an alert is triggered

Updates Every 5 Seconds

Data refreshes automatically throughout the entire lecture

Technologies Used

FastAPI

Fast Python framework that serves the deep learning model and responds to web requests in real time.

Deep Learning

A neural network trained to detect eye gaze direction so classify focus level from camera images.

PHP

PHP handles the website logic and user management

MySQL

Stores all session data, focus scores, lecture records, and user accounts securely.

System Workflow

01

Student joins lecture

Logs in & opens the monitoring page with webcam access

02

Camera captures frames

JavaScript sends a snapshot to the server every few seconds

03

FastAPI receives image

The AI receives the image and runs it through the model and inference : Focused / No Face

04

Model detect gaze

Deep learning model returns the yaw and pitch numbers

05

Score saved to database

MySQL stores the result with timestamp and lecture ID

06

Teacher dashboard updates

Teacher sees live focus stats and post-session reports

03

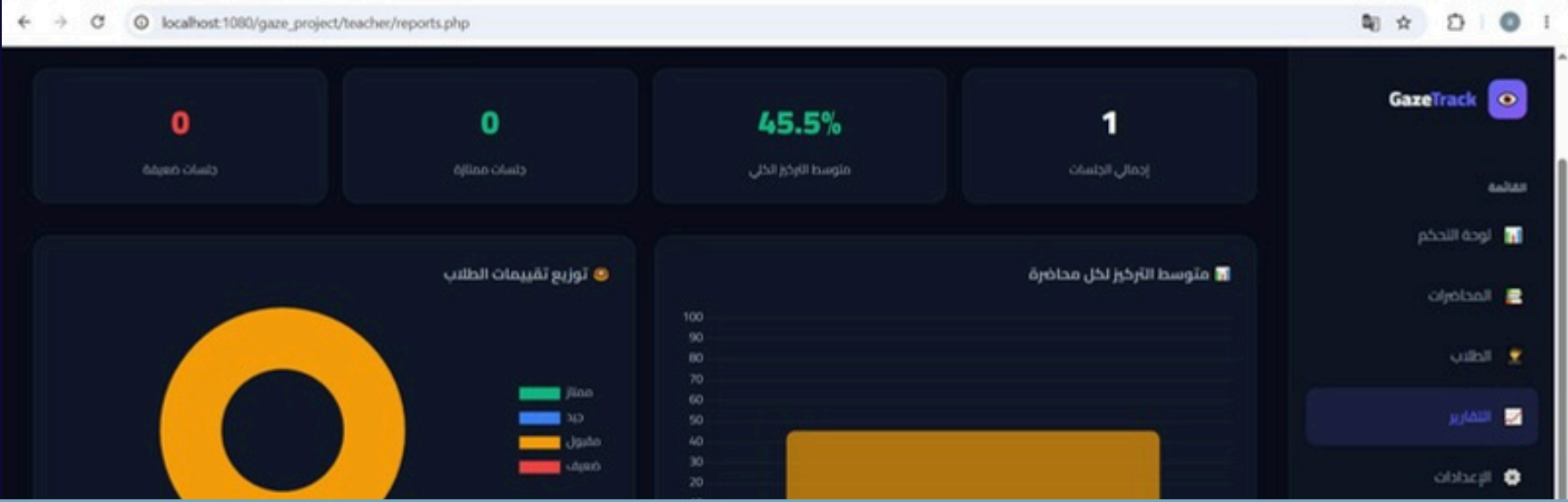
Alert



**If a student is distracted more than 3 times
An alert appears with their name and focus score**

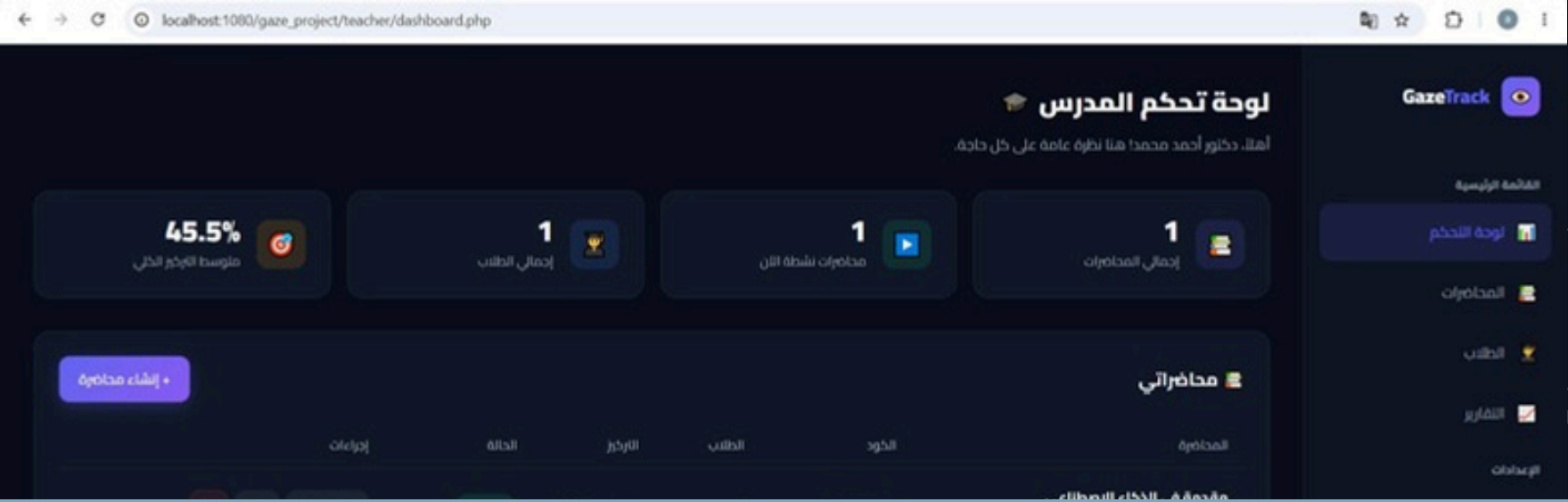
System Interfaces

Teacher Dashboard

The screenshot shows the Teacher Dashboard interface. At the top, there's a header with the GazeTrack logo. Below it, a row of four summary cards displays: 0 Lectures, 0 Students, 45.5% Average Focus, and 1 Total Sessions. The main area features a donut chart titled 'Distribution of Student Ratings' with a legend for Excellent (green), Good (blue), Average (yellow), and Poor (red). To the right is a bar chart titled 'Average Focus per Lecture'. A sidebar on the right contains navigation links like 'Dashboard', 'Reports', 'Students', 'Lectures', 'Settings', and 'Logout'.

Overview of lectures, students & focus averages

Teacher Reports Page

The screenshot displays the Teacher Reports Page. It features a header with the GazeTrack logo and a title 'Teacher Reports Page'. The main content area includes a row of four summary cards: 45.5% Average Focus, 1 Total Sessions, 1 Active Sessions, and 1 Total Sessions. Below this is a section titled 'My Lectures' with a table showing lecture details. A sidebar on the right contains navigation links like 'Dashboard', 'Reports', 'Students', 'Lectures', 'Settings', and 'Logout'.

Detailed charts, ratings & session history

Student Monitor

The screenshot shows the Student Monitor interface. It features a header with the GazeTrack logo and a title 'Student Monitor'. The main content area includes a row of four summary cards: 0 Focus, 0 Students, 0 Average Focus, and 0 Total Sessions. Below this is a section titled 'My Progress' with a table showing progress details. A sidebar on the right contains navigation links like 'Dashboard', 'Reports', 'Students', 'Lectures', 'Settings', and 'Logout'.

Live camera feed with real-time focus tracking

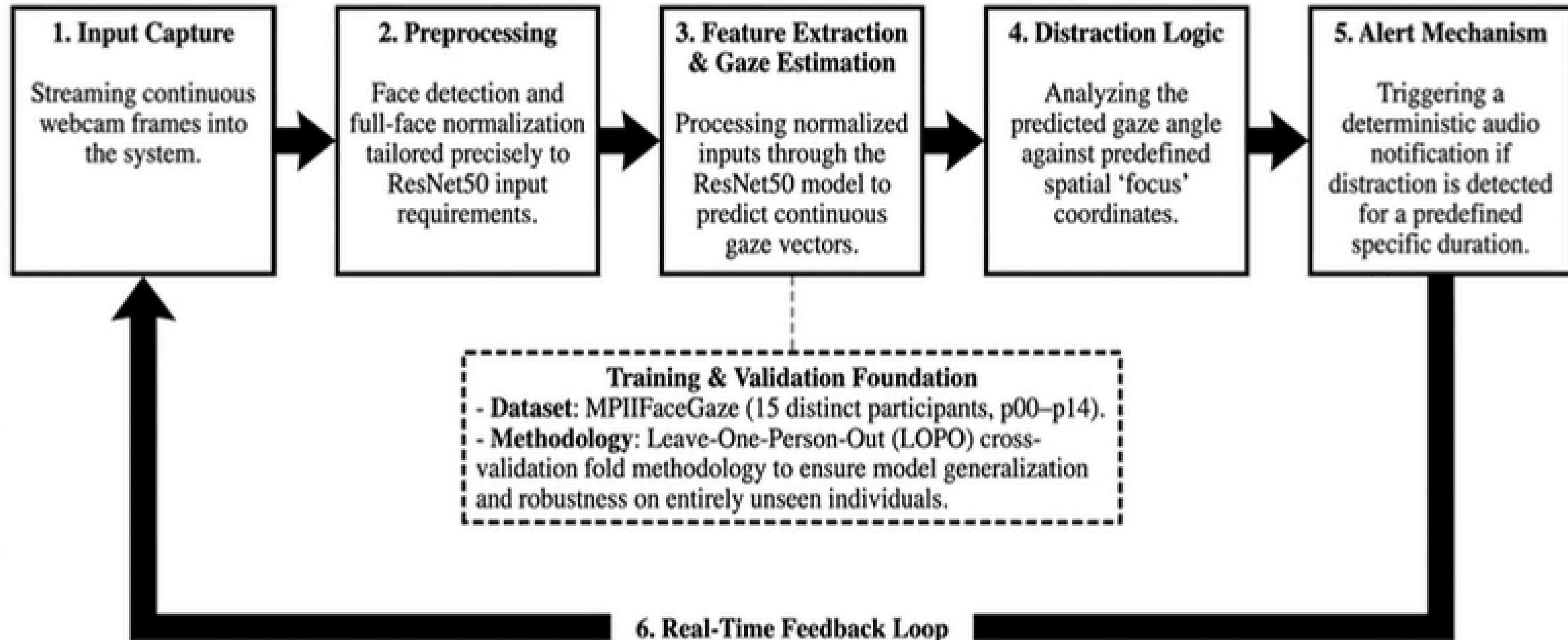
Student Progress

The screenshot displays the Student Progress interface. It features a header with the GazeTrack logo and a title 'Student Progress'. The main content area includes a row of four summary cards: 0 Focus, 45.5% Average Focus, 45.5% Average Focus, and 1 Total Sessions. Below this is a section titled 'My Progress' with a table showing progress details. A sidebar on the right contains navigation links like 'Dashboard', 'Reports', 'Students', 'Lectures', 'Settings', and 'Logout'.

Personal stats, rating distribution & history

Conclusion

System Algorithm & Logical Workflow

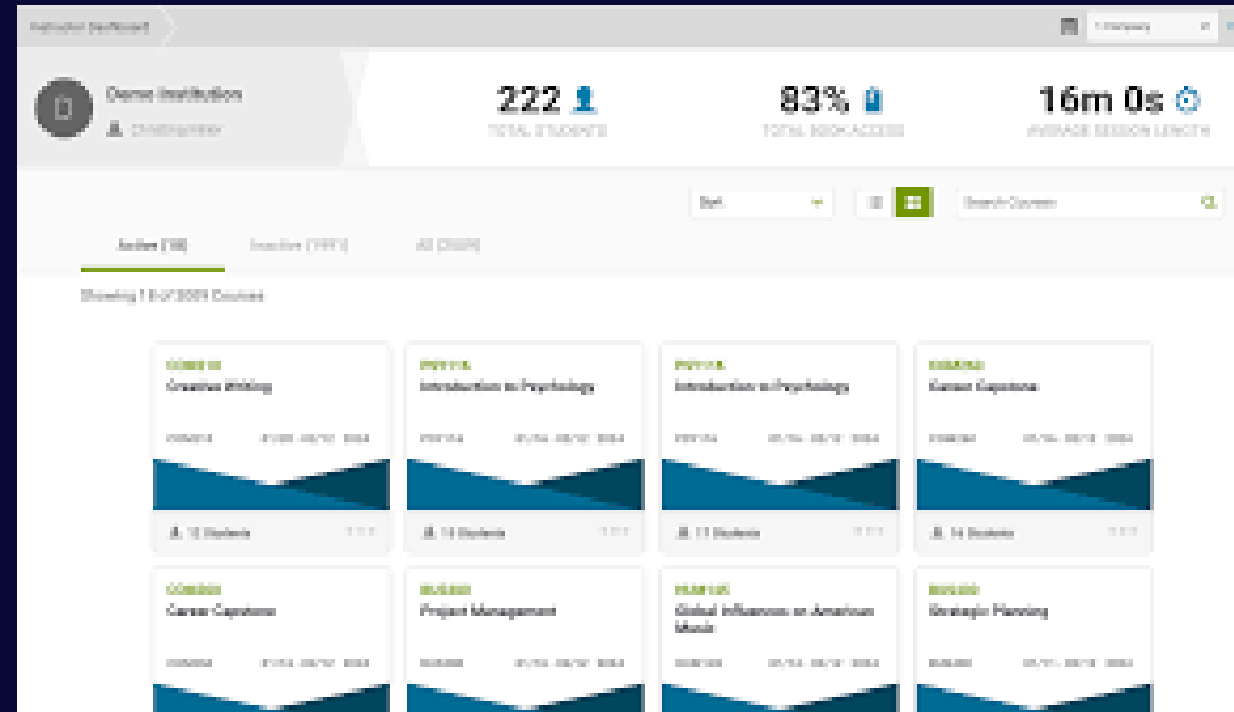


Future Work

Gaze Heatmap



Instructor Dashboard



Employee productivity measurement

26 Presentation



Thank You

